

Solvent Extraction

The techniques of solvent extraction and solid-phase extraction and related techniques are very useful for isolating analytes from complex sample matrices prior to chromatographic analysis. Solvent extraction is also useful for spectrophotometric determination.

Solvent extraction involves the distribution of a solute between two immiscible liquid phases. This technique is extremely useful for very rapid and “clean” separations of both organic and inorganic substances. In this chapter, we discuss the distribution of substances between two phases and how this can be used to form analytical separations.

The solvent extraction of metal ions into organic solvents is described. Solid-phase extraction is a technique in which hydrophobic functional groups are bonded to solid particle surfaces and act as the extracting phase. They reduce the need for large volumes of organic solvents.

Distribution Coefficient

A solute S will distribute itself between two phases (after shaking and allowing the phases to separate) and, within limits, the ratio of the concentrations of the solute in the two phases will be a constant:

$$K_D = \frac{[S]_1}{[S]_2} \quad (18.1)$$

Where K_D is the **distribution coefficient** and the subscripts represent solvent 1 (e.g., an organic solvent) and solvent 2 (e.g., water). If the distribution coefficient is large, the solute will tend to be quantitatively partitioned in solvent 1.

The apparatus used for solvent extraction is the separatory funnel, illustrated water into organic solvents. “Like dissolves like.” in Figure 18.1. Most often, a solute is extracted from an aqueous solution into an immiscible organic solvent. After the mixture is shaken for about a minute, the phases are

allowed to separate and the bottom layer (the denser solvent) is drawn off in a completion of the separation.



Fig. 18.1. Separatory funnel.

Many substances are partially ionized in the aqueous layer as weak acids. The extraction now becomes dependent on the pH of the solution. Consider, for example, the extraction of benzoic acid from an aqueous solution into ether. Benzoic acid (HBz) is a weak acid in water with a particular ionization constant K_a (given by Equation 18.4). The distribution coefficient is given by

$$K_D = \frac{[\text{HBz}]_e}{[\text{HBz}]_a} \quad (18.2)$$

where e represents ether and a represents the aqueous solvent. However, part of the benzoic acid in the aqueous layer will exist as Bz^- that does not transfer into ether, depending on the magnitude of K_a and on the pH of the aqueous layer; hence, quantitative separation may not be achieved when a significant fraction exists as Bz^- .

18.2 Distribution Ratio

It is more meaningful to describe a different term, the **distribution ratio**, which is the ratio of the concentrations of *all* the species of the solute in each phase. In this example, it is given by

$$D = \frac{[\text{HBz}]_e}{[\text{HBz}]_a + [\text{Bz}^-]_a} \quad (18.3)$$

We can readily derive the relationship between **D** and **KD** from the equilibria involved. The acidity constant **K_a** for the ionization of the acid in the aqueous phase is given by

$$K_a = \frac{[\text{H}^+]_a [\text{Bz}^-]_a}{[\text{HBz}]_a} \quad (18.4)$$

Hence,

$$[\text{Bz}^-]_a = \frac{K_a [\text{HBz}]_a}{[\text{H}^+]_a} \quad (18.5)$$

From Equation 18.2,

$$[\text{HBz}]_e = K_D [\text{HBz}]_a \quad (18.6)$$

Substitution of Equations 18.5 and 18.6 into Equation 18.3 gives

$$D = \frac{K_D [\text{HBz}]_a}{[\text{HBz}]_a + K_a [\text{HBz}]_a / [\text{H}^+]_a} \quad (18.7)$$

$$D = \frac{K_D}{1 + K_a / [\text{H}^+]_a} \quad (18.8)$$

This equation predicts that when $[\text{H}^+]_a \gg K_a$, D is nearly equal to K_D , and if K_D is large, the benzoic acid will be quantitatively extracted into the ether layer; D is maximum under these conditions. If, on the other hand, $[\text{H}^+]_a \ll K_a$, then D reduces to $K_D [\text{H}^+]_a / K_a$, which will be small, and the benzoic acid will remain in the aqueous layer. That is, in alkaline solution, the benzoic acid is ionized and cannot be extracted, while in acid solution, it is largely undissociated.

These conclusions are what we would intuitively expect from inspection of the chemical equilibria.

Equation 18.8, like Equation 18.1, predicts that the extraction efficiency will be independent of the original concentration of the solute. This is one of the attractive features of solvent extraction; it is applicable to tracer (e.g., radioactive) levels and to macro levels alike, a condition that applies only so long as the solubility of the solute in one of the phases is not exceeded and there are no side reactions such as dimerization of the extracted solute.

Of course, if the hydrogen ion concentration changes, the extraction efficiency (**D**) will change. In this example, the hydrogen ion concentration will increase with increasing benzoic acid concentration, unless an acid–base buffer is added to maintain the hydrogen ion concentration constant (see Chapter 7 for a discussion of buffers).

In deriving Equation 18.8, we actually neglected to include in the numerator of Equation 18.3 a term for a portion of the benzoic acid that exists as the dimer in the organic phase. The extent of dimerization tends to increase with increased concentration, and by **Le Chatelier's principle**, this will cause the equilibrium to shift in favor of the organic phase with increased concentration. So, in cases such as this, the efficiency of extraction will actually increase at higher concentrations. As an exercise, derivation of the more complete equation is presented in Problem 12.

18.3 Percent Extracted

The distribution ratio **D** is a constant independent of the volume ratio. However, the fraction of the solute extracted will depend on the volume ratio of the two solvents. If a larger volume of organic solvent is used, more solute must dissolve in this layer to keep the concentration ratio constant and to satisfy the distribution ratio.

The fraction of solute extracted is equal to the millimoles of solute in the organic layer divided by the total number of millimoles of solute. The millimoles are given by the molarity times the milliliters. Thus, the percent extracted is given by;

$$\%E = \frac{[S]_o V_o}{[S]_o V_o + [S]_a V_a} \times 100\% \quad (18.9)$$

where **V_o** and **V_a** are the volumes of the organic and aqueous phases, respectively. It can be shown from this equation (see Problem 11) that the percent extracted is related to the distribution ratio by;

$$\%E = \frac{100D}{D + (V_a/V_o)} \quad (18.10)$$

If $V_a = V_o$, then

$$\%E = \frac{100D}{D + 1} \quad (18.11)$$

In the case of equal volumes, the solute can be considered quantitatively retained in the aqueous phase if **D** is less than 0.001. It is essentially quantitatively extracted if **D** is greater than 1000. The percent extracted changes only from 99.5 to 99.9% when **D** is increased from 200 to 1000.

Solvent Extraction of Metals

Solvent extraction has one of its most important applications in the separation of metal cations. In this technique, the metal ion, through appropriate chemistry, distributes from an aqueous phase into a water-immiscible organic phase. Solvent extraction of metal ions is useful for removing them from an interfering matrix, or for selectively (with the right chemistry) separating one or a group of metals from others.

The separation can be accomplished in several ways. You have noted above that the uncharged organic molecules tend to dissolve in the organic layer while the charged anion from the ionized molecules remains in the polar aqueous layer. This is an example of “like dissolves like.” Metal ions do not tend to dissolve appreciably in the organic layer. For them to become soluble, their charge must be neutralized and something must be added to make them organic-like. There are two principal ways of doing this.

EXTRACTION OF ION-ASSOCIATION COMPLEXES

In one method, the metal ion is incorporated into a bulky molecule and then associates with another ion of the opposite charge to form an ion pair, or the metal ion associates with another ion of great size (organic-like). For example, it is well known that iron (III) can be quantitatively extracted from hydrochloric acid medium into diethyl ether. The mechanism is not completely understood, but evidence exists that the chloro complex of the iron is coordinated with the oxygen atom of the solvent (the solvent displaces the coordinated water), and this ion associates with a solvent molecule that is coordinated with a proton:



Similarly, the uranyl ion UO_2^{2+} is extracted from aqueous nitrate solution into isobutanol by associating with two nitrate ions (UO_2^{2+} , 2NO_3^-), with the uranium probably being solvated by the solvent to make it solvent-like. Permanganate forms an ion pair with tetraphenylarsonium ion [$(\text{C}_6\text{H}_5)_4\text{As}^+$, MnO_4^-], which makes it organic-like, and it is extracted into methylene chloride. There are numerous other examples of ion-association extractions. Unsightly foam often appears in the shoreline of natural waters, sometimes because of the presence of natural surfactants but also often because of contamination from man-made detergents. These are mostly anionic, and anionic surfactants can be extracted into an organic solvent such as chloroform as the ion pair with the cationic dye methylene blue.

Halogenated organic solvents have often been used in the past as extraction solvents of choice. There is considerable concern about their toxicity, however, and in most laboratories the use of chlorinated organic solvents are being phased out. These solvents tend to be more polar than hydrocarbon solvents and can sometimes be replaced by water-insoluble ionic liquids.

EXTRACTION OF METAL CHELATES

The most widely used method of extracting metal ions is formation of a chelate molecule with an organic chelating agent. As mentioned in Chapter 9, a chelating agent contains two or more complexing groups. Many of these reagents form colored chelates with metal ions and form the basis of

spectrophotometric methods for determining the metals. The chelates are often insoluble in water and will precipitate. They are, however, usually soluble in organic solvents such as methylene chloride. Many of the organic precipitating agents listed in Chapter 10 are used as extracting agents.

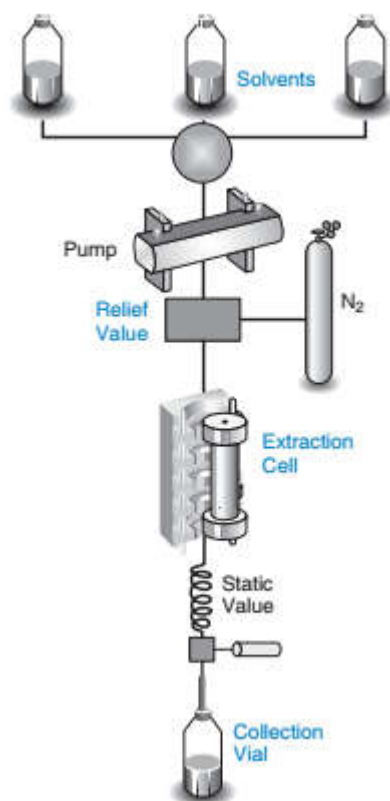
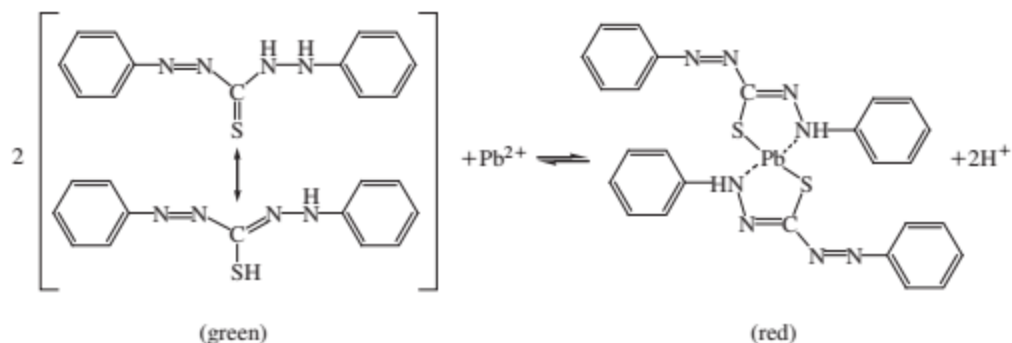


Figure 18.2. Schematic depiction of operation of an Accelerated Solvent Extraction Instrument.

EXTRACTION PROCESS FOR METAL CHELATES

Most chelating agents are weak acids that ionize in water; the ionizable proton is displaced by the metal ion when the chelate is formed, and the charge on the organic compound neutralizes the charge on the metal ion. An example is diphenylthiocarbazone (dithizone), which forms a chelate with lead ion:



The usual practice is to add the chelating agent, HR, to the organic phase. It distributes between the two phases, and in the aqueous phase it dissociates as a weak acid. The metal ion, M^{n+} , reacts with nR^- to form the chelate MR_n , which then distributes into the organic phase. The distribution ratio is given by the ratio of the metal chelate concentration in the organic phase to the metal ion concentration in the aqueous phase. The following equation can be derived:

$$D = \frac{[MR_n]_o}{[M^{n+}]_a} = K \frac{[HR]_o^n}{[H^+]_a^n} \quad (18.12)$$

where K is a constant that includes K_a of HR, K_f of MR_n , and K_D of HR and MR_n . Note that the distribution ratio is independent of the concentration of the metal ion, provided the solubility of the metal chelate in the organic phase is not exceeded. HR is often in large excess and is considered constant. The extraction efficiency can be affected only by changing the pH or the reagent concentration. A 10-fold increase in the reagent concentration will increase the extraction efficiency the same as an increase in the pH of one unit (10-fold decrease in $[H^+]$). Each effect is greater as n becomes greater. By using a high concentration of reagent, extraction can be performed in more acid solution.

Chelates of different metals extract at different pH values. Some can be extracted over a broad range of pH, whereas others may only be extracted from alkaline solution. Once a metal has been extracted into an organic solvent and the solvent separated, it may be extracted back into an aqueous layer, if desired, at an appropriate lower pH. Such a process is often referred to as back extraction. By appropriate adjustment of pH, selectivity can be

achieved in the extraction. Also, judicious use of masking agents, complexing agents that prevent one metal ion from reacting with the chelating agent, can enhance the selectivity.

For reasons of toxicity and disposal, most organic solvent extraction methods for metals have been replaced by solid-phase resins bearing chelating functionalities. Alternatively, ICP-based methods are used; these have high sensitivities and few interferences and do not require extraction, as described in Chapter 17. The principles of solvent extraction, especially sequential extraction, are key to understanding how chromatography (next chapter) works, essentially by the same continuous sequential partitioning process.